

Satyajit Ghana

Head of Engineering · 3D Vision · MLOps

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SUMMARY

Engineering leader and deep-learning engineer specializing in 3D computer vision, SLAM, and production MLOps. I design end-to-end systems — from LiDAR point-cloud reconstruction and custom neural networks to scalable cloud deployment — and lead teams shipping construction-AI scanning products. I also design and teach production MLOps curricula.

EXPERIENCE

Head of Engineering

Apr 2023 – Present

Inkers Technology Pvt Ltd, Bangalore

- Led development of new laser-scanning device prototypes with improved laser accuracy and integrated RGB and thermal cameras.
- Developed a modified SLAM algorithm that enables scanning in small, confined spaces and reduced drift by 50%.
- Built a point-cloud cleaning algorithm that reduced cloud thickness from 2 cm to 1 cm while preserving detail.

MLOps Instructor

Feb 2022 – Present

The School of AI, Bangalore

- Designed and taught the EMLOV2 and EMLOV3 production-MLOps courses end to end.
- Covered containerized ML, Kubernetes/EKS model serving, MLOps pipelines, and LLM fine-tuning (PEFT/LoRA, vLLM, GGML & llama.cpp, LlamaIndex & LangChain).

Deep Learning Software Engineer

Apr 2022 – May 2023

Inkers Technology Pvt Ltd, Bangalore

- Optimized the point-cloud cleaning and meshing algorithm to produce meshes accurate to within 1 cm.
- Designed and trained a custom transformer for 3D mesh part segmentation, deriving construction material quantities.
- Reduced mesh sizes from 1.2 GB to 120 MB via texture optimization, compression, and Blender decimation, enabling in-browser rendering.

Associate Deep Learning Software Engineer

Jul 2021 – Apr 2022

Inkers Technology Pvt Ltd, Bangalore

- Developed the full-stack web app for Observance.AI, an end-to-end construction-AI platform.
- Migrated facial-recognition onboarding to AWS Lambda, onboarding people in minutes with up to 2,000 parallel Lambdas.
- Optimized the data pipeline on HPC instances (p3/p4, g4dn, r6i), processing 1–2 TB/day on S3 with 128-core / 1024 GB spot instances.

Visual Processing Intern

Jun 2020 – Jul 2021

Inkers Technology Pvt Ltd, Bangalore

- Designed a custom GStreamer plugin for DeepStream performing near-real-time stereo rectification, and deployed on edge devices (Xavier NX) via Docker at stereo 4K @ 30 fps for detection and tracking.
- Optimized YOLOv4-Tiny for TensorRT for a 2x speedup and managed multi-GPU training on AWS (8x V100) with data backup and synchronization to cut costs.

SKILLS

Languages | Python, C/C++, JavaScript, TypeScript, Java, Haskell, Go

ML & Deep Learning | PyTorch, PyTorch3D, HuggingFace, TensorRT, DeepStream, vLLM, PEFT/LoRA, Stable Diffusion/SDXL, LlamaIndex, LangChain

3D & Vision | SLAM, 3D Reconstruction, Point Clouds (PCL, Open3D, CloudCompare, MeshLab), Blender (BPY), THREE.js, OpenCV

MLOps & Cloud | Docker, Kubernetes (EKS/ECS), KServe, Kubeflow, TorchServe, ISTIO, AWS (Lambda, SageMaker, S3, Kinesis, EC2, ECR, EFS), Azure, Prometheus, Grafana

Backend & Systems | FastAPI, gRPC, Redis, Kafka, RabbitMQ, WebRTC, ROS1, Hydra

Frontend & Mobile | React, TailwindCSS, Flutter

EDUCATION

B.Tech, Computer Science & Engineering

Ramaiah University of Applied Sciences, Bangalore

Aug 2017 – Mar 2021

CGPA 9.78/10 • Silver Medalist